

Example: Document vision pack for brief 3, nameplate and diagram capture

What this is: the vision pack filled in for use case brief 3, the way a group might write it at the end of Day 3. The scores are from labs 08 and 09 in the repository. It shows the level of detail we are after. It is one answer, not the answer, and it leaves real gaps open on purpose. Synthetic program material: no real OQ data.

Field	Answer
Author	Example group (brief 3)
System	Brief 3: nameplate and diagram capture
Date	Day 3, 29 September 2026

1. What images, and where they may go

Image type	How many a month	Classification	May it leave the tenant?	Self-hosted or vendor API
Nameplate photos from walk-downs	About 400	Internal	Yes, through a tenant deployment	Vendor model deployed in the tenant
Scanned work order cards	About 150	Internal, names of staff	Yes, through a tenant deployment	Vendor model deployed in the tenant
P&IDs and process diagrams	About 30	Restricted	No	Self-hosted on a GPU VM

2. The extraction contract

Field	Type	Required?	Null allowed?	Checked against
equipment_tag	Text, tag pattern	Yes	Yes, then to a person	Equipment master (AssetHive, read tool)
manufacturer	Text	Yes	Yes	Approved manufacturer list
model_number	Text	Yes	Yes	Equipment master
serial_number	Text	Yes	Yes	Equipment master; never auto-accepted
rating (kW, barg)	Number and unit	No	Yes	Range for the equipment class
permit_number (work orders)	Text, PTW pattern	For hands-on work	Yes, then to a person	Permit register

3. The test set, in tiers

Tier	What it is	How many images	Where they come from
1 Clean	Digital forms and clean diagrams	3 + 3	Labs 08 and 09
2 Degraded	Office scans; dense diagrams	3 + 3	Labs 08 and 09
3 Worst real	Faxed with a stamp;	3 + 3	Labs 08 and 09

Tier	What it is	How many images	Where they come from
	diagrams photographed from a printout		
Known-bad	wo_10 with three blank fields; dia_10 with a line crossing and P-101A beside P-101B	1 + 1	Labs 08 and 09

Gap: none of these are nameplates. Twenty real walk-down photos, with glare, are the next test set. Not collected yet.

4. Scores, never folded together

Tier	Model	Correct	Missed	Invented	Score
Forms, tier 1	gpt-4.1-mini	54 of 54 fields	0	0	1.000
Forms, tier 2	gpt-4.1-mini	53 of 54	0	0	0.981
Forms, tier 3 (fax)	gpt-4.1-mini	46 of 54	0	0	0.852
Forms, known-bad	gpt-4.1-mini	18 of 18	0	0	1.000
Diagrams, tiers 1 to 3	gpt-4.1-mini	nodes all right; arrow labels 0.8 on photos		0 invented arrows	1.000 / 0.979 / 0.979
Forms and diagrams	qwen2.5vl:3b	measured live in the room		at least one invented value in a verified run	

Scores are the notebooks' saved runs. The small self-hosted model is the one that invents; the exact count moves between runs, the direction does not.

5. Validation and confidence

Check	Your rule	What happens when it fails
Format	Tag, serial and permit patterns; dates as YYYY-MM-DD	To a person
Lookup in the register	The tag must exist in the equipment master	Refused, never written
Date and sequence logic	Sign-off not before raised date	To a person
Business rule	Hands-on work needs a permit number	To a person
Confidence below threshold	Any field below 0.8 (Document Intelligence confidence)	To a person
Any invented or null required field	Never auto-accepted	To a person, with the image

The lab 09 lesson: format rules caught none of the 4 wrong records. The register lookup and the person are what catch a wrong but well-formed value.

6. The human gate

- Who confirms, what do they see, how long may it wait?

The maintenance planner for the area. They see the photo beside the proposed record and the current equipment master record, with differences highlighted. A proposal waits up to 5 working days, then expires.

- Which fields are never written without a person?

Serial number, model number and anything that changes a spare-parts link. In version 1 that is every write: nothing is written to the equipment master without a person.

7. Into the index

- Marked as machine-read, with the image ID?

Yes. Every extracted record carries source = extraction, the model, the date and the image ID, and answers say "read from a photo" when they use one.

- Which fields are re-read from the pixels?

Permit number and serial number, whenever an answer that uses them would start work or order a part.

8. Retention

- Where, how long, who can see?

Blob Storage in the tenant, 3 years, maintenance planning and reliability only. Photos with people in them are deleted after extraction. Retention period not yet agreed with records management.

9. In Azure

Piece	In the lab	In Azure	Your choice
Vendor vision model	gpt-4.1-mini	Foundry deployment	gpt-4.1-mini for nameplates and cards
Self-hosted vision model	qwen2.5vl:3b on a T4	GPU VM in the tenant	For P&IDs only, and only after it passes the tier test
Forms and scans	Schema and prompt	Document Intelligence	Custom model for the work order card; its per-field confidence drives section 5
Into the index	Notebook index	Azure AI Search	Extracted records beside the text, marked machine-read
Human gate	A flag	Review queue	A queue in the maintenance planning tool

10. If someone pitches industrial computer vision

Item	Your answer
Use case, in one sentence	Out of scope for brief 3. Recorded here because a vendor offered PPE detection at the same meeting
Cameras	Not scoped
Data	Not scoped
Success metric	Not scoped
Owners	OT would lead; IT runs the pipeline
Stop criteria	Not scoped

Five questions for the vendor, asked at that meeting.

- On whose images was it trained?

"Many industrial sites." No number, and none from desert sites. Follow up.

- Miss and false alarm rates on our footage, measured by us?

Not offered. Condition for any pilot.

- Where does the video go?

Their cloud by default. Would fail the Day 1 gate.

- Camera moves or lighting changes?

"The model adapts." No retraining loop described.

- Cost per camera per year?

Not given.